CLOUD COMPUTING

ASSIGNMENT – 1

Submitted By:-

R PRANAV

MCA S2

ROLL NO:46

Hosting a Custom Website on AWS EC2 using Apache

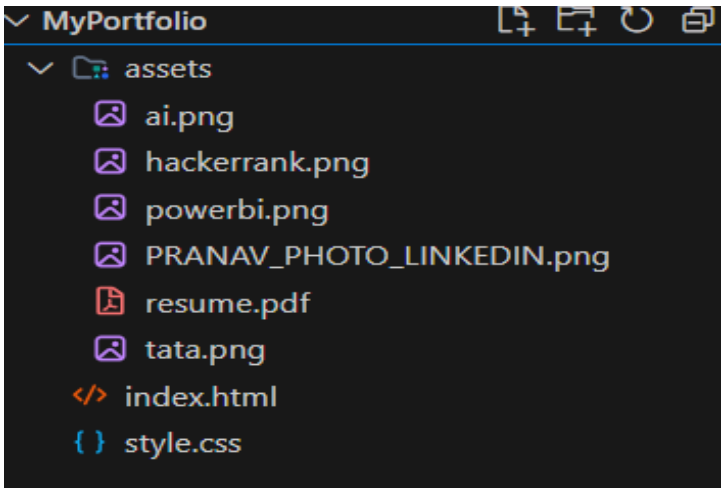
1)Aim

To host a custom website on an Ubuntu EC2 instance using Apache Web Server and make it accessible via public IPv4.

2) Requirements

- AWS Account (Amazon Web Services)
- Ubuntu EC2 Instance
- Apache Web Server
- Portfolio Website (HTML, CSS, assets)
- SSH client

3) Project Structure



4) Procedure

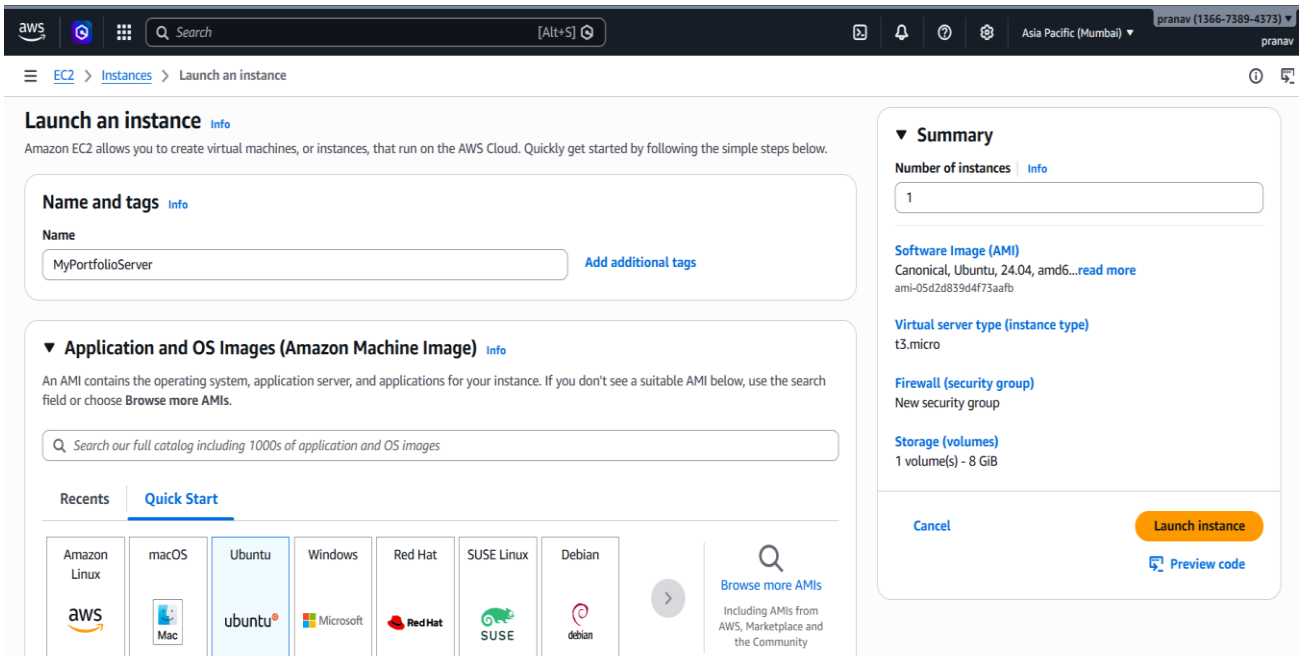
Step 1: Launch EC2 Instance

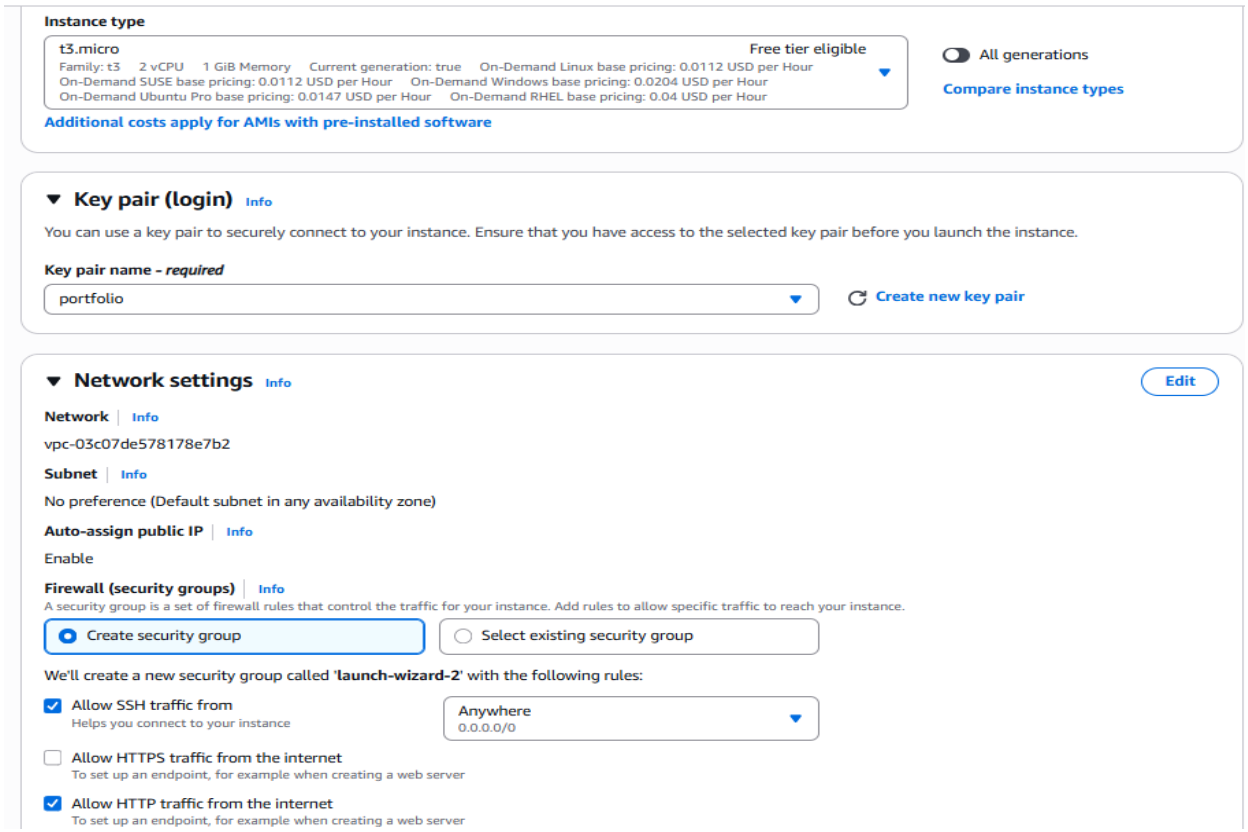
- Login to AWS Console
- Go to **EC2 Dashboard**
- Click **Launch Instance**

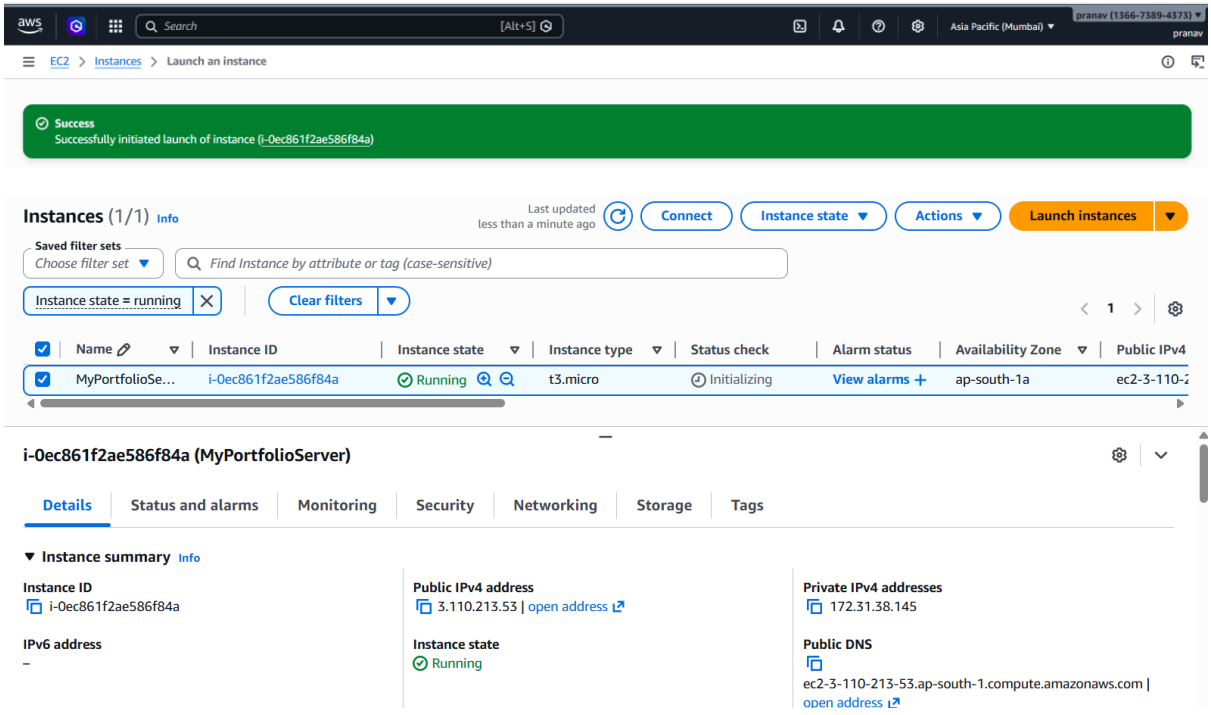
Configure:

- Name: MyPortfolioServer

- AMI: Ubuntu Server 22.04
- Instance Type: t3.micro
- Key Pair: Create/download .pem
- Network Settings:
    - Allow **SSH (22)**
    - Allow **HTTP (80)**







Step 2: Connect to Instance (SSH)

Open terminal:

```
ssh -i "C:\Users\LENOVO\Downloads\portfolio.pem" ubuntu@3.110.213.53
```

```
C:\Users\LENOVO>ssh -i "C:\Users\LENOVO\Downloads\portfolio.pem" ubuntu@3.110.213.53
The authenticity of host '3.110.213.53 (3.110.213.53)' can't be established.
ED25519 key fingerprint is SHA256:R8j57kcp/+tEEkrhIVSORCdMZcDvzMf/tOu8qLdUaWs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.110.213.53' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Mar 26 14:28:13 UTC 2026

  System load:  0.25              Temperature:           -273.1 C
  Usage of /:   26.4% of 6.71GB   Processes:             113
  Memory usage: 26%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.38.145

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

Step 3: Install Apache Web Server

Run:

```
sudo apt update
sudo apt install apache2 -y
```

Start Apache:

```
sudo systemctl start apache2
sudo systemctl enable apache2
```

```
ubuntu@ip-172-31-38-145:~$ sudo apt update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1847 kB]
Get:14 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:15 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:16 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:17 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1665 kB]
Get:18 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [323 kB]
Get:19 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:20 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.2 kB]
Get:21 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2837 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1551 kB]
Get:23 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [656 kB]
Get:24 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:25 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.2 kB]
Get:26 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7520 B]
Get:27 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:28 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [500 B]
Get:29 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:30 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9208 B]
Get:31 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7344 B]
Get:32 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
```

```
ubuntu@ip-172-31-38-145:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 26 not upgraded.
Need to get 2091 kB of archives.
After this operation, 8113 kB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Progress: [ 98%] [#################################################################################################################
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.11 [1335 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.11 [163 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.11 [98.2 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.11 [90.2 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 2091 kB in 0s (48.6 MB/s)
Preconfiguring packages ...
```

```
ubuntu@ip-172-31-38-145:~$ sudo systemctl start apache2
ubuntu@ip-172-31-38-145:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
```

Step 4: Verify Apache Installation

Open browser:

http:// 3.110.213.53

You should see Apache default page.

Step 5: Upload Website Files

The website files were transferred securely from the local system to the EC2 instance.

Open Command Prompt

Navigate to your project directory:

```
cd "C:\Users\LENOVO\OneDrive\Desktop"
```

Execute SCP command:

```
scp -i "C:\Users\LENOVO\Downloads\portfolio.pem" -r
"C:\Users\LENOVO\OneDrive\Desktop\MyPortfolio" ubuntu@3.110.213.53:/home/ubuntu
```

```
C:\Users\LENOVO\OneDrive\Desktop>scp -i "C:\Users\LENOVO\Downloads\portfolio.pem" -r "C:\Users\LENOVO\OneDrive\Desktop\MyPortfolio" ubuntu@3.110.213.53:/home/ubuntu
COMMIT_EDITMSG                                                                                                                    100%   28     0.7KB/s   00:00
config                                                                                                                            100%  313     8.0KB/s   00:00
description                                                                                                                       100%   73     1.9KB/s   00:00
FETCH_HEAD                                                                                                                        100%  114     2.8KB/s   00:00
HEAD                                                                                                                              100%   21     0.6KB/s   00:00
applypatch-msg.sample                                                                                                             100%  478    11.1KB/s   00:00
commit-msg.sample                                                                                                                 100%  896    26.5KB/s   00:00
fsmonitor-watchman.sample                                                                                                         100% 4726    82.4KB/s   00:00
post-update.sample                                                                                                                100%  189     5.3KB/s   00:00
pre-applypatch.sample                                                                                                             100%  424    11.8KB/s   00:00
pre-commit.sample                                                                                                                 100% 1649    38.3KB/s   00:00
pre-merge-commit.sample                                                                                                           100%  416    11.0KB/s   00:00
pre-push.sample                                                                                                                   100% 1374    34.4KB/s   00:00
pre-rebase.sample                                                                                                                 100% 4898   125.9KB/s   00:00
pre-receive.sample                                                                                                                100%  544    14.0KB/s   00:00
prepare-commit-msg.sample                                                                                                         100% 1492    35.5KB/s   00:00
push-to-checkout.sample                                                                                                           100% 2783    69.7KB/s   00:00
sendemail-validate.sample                                                                                                         100% 2308    55.0KB/s   00:00
update.sample                                                                                                                     100% 3650    89.1KB/s   00:00
index                                                                                                                             100%  835    25.5KB/s   00:00
exclude                                                                                                                           100%  240     6.5KB/s   00:00
HEAD                                                                                                                              100% 1300    27.0KB/s   00:00
main                                                                                                                              100% 1300    28.9KB/s   00:00
main                                                                                                                              100% 1108    27.1KB/s   00:00
28d835bbf2bd3cb1a72a15878904b784b91751                                                                                            100%  228     6.4KB/s   00:00
23d66bb41083712c19ed5be1cb6baa048975e3                                                                                            100%  122     2.5KB/s   00:00
ba08fdc71513e6621710ec1a1a3dffa764d4c1                                                                                            100%   62KB 602.2KB/s   00:00
049a27cc5b070caa4364c7967ff227f8b17dbd                                                                                            100% 6280   153.3KB/s   00:00
370ddc4a62086bcc520db3f103c3d46c524a9b                                                                                            100%  121     2.4KB/s   00:00
6fbae6f87528f554a86202d253379a4851145b                                                                                            100%   62KB 766.0KB/s   00:00
16a507096ea22e63816b32e61a9c5700675909                                                                                            100% 6442   174.8KB/s   00:00
425395bff88bd9ba73bf54f03e28ba6cb0d7ba                                                                                            100%  228     7.2KB/s   00:00
ab01cdd4fcb38128b79cca558c634660f23aa0                                                                                            100%  228     6.0KB/s   00:00
4e782d51a79e692fd073d456c108fc1b8a64b5                                                                                            100%  219     4.4KB/s   00:00
ac550a0be919a58c02ff0bdee9572ae7f45d45                                                                                            100%  177     4.4KB/s   00:00
6415439099e54006500a32662b8f127ba54f22                                                                                            100%  142     3.8KB/s   00:00
e9d2a81e6a46bcb2687de135fb99bc66a6e435                                                                                            100%  189KB   1.0MB/s   00:00
f03b24d4627f0b072585404e5994cfb89a207b                                                                                            100% 6115   138.9KB/s   00:00
029253b124af7202ce3fb52278be69ec1e2a14                                                                                            100%  121     1.9KB/s   00:00
b49ccb02e29e5a35322062a387a9b4b5181189                                                                                            100%   77KB 924.9KB/s   00:00
34a2fd86bbba207661cce7046b6db37240f7ab                                                                                            100%  182KB   1.8MB/s   00:00
```

Step 6: Move Files to Apache Web Directory

After successfully uploading the project using SCP, connect to the EC2 instance using SSH:

```
ssh -i "C:\Users\LENOVO\Downloads\portfolio.pem" ubuntu@3.110.213.53
```

```
C:\Users\LENOVO\OneDrive\Desktop>ssh -i "C:\Users\LENOVO\Downloads\portfolio.pem" ubuntu@3.110.213.53
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Mar 26 14:54:06 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   30.4% of 6.71GB   Processes:             119
  Memory usage: 27%               Users logged in:       1
  Swap usage:   0%                IPv4 address for ens5: 172.31.38.145


Expanded Security Maintenance for Applications is not enabled.

32 updates can be applied immediately.
31 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Thu Mar 26 14:36:17 2026 from 13.233.177.4
```

Navigate to the uploaded project folder:

```
cd /home/ubuntu/MyPortfolio
```

Copy website files to Apache root directory:

```
sudo cp -r * /var/www/html/
```

Remove default Apache index file (if present):

```
sudo rm /var/www/html/index.html
```

```
ubuntu@ip-172-31-38-145:~$ cd /home/ubuntu/MyPortfolio
ubuntu@ip-172-31-38-145:~/MyPortfolio$ sudo cp -r * /var/www/html/
ubuntu@ip-172-31-38-145:~/MyPortfolio$ sudo rm /var/www/html/index.html
ubuntu@ip-172-31-38-145:~/MyPortfolio$
```

Step 7: Set File Permissions

Assign proper permissions to ensure the web server can access the files:

```
sudo chmod -R 755 /var/www/html
```

Step 8: Verify Uploaded Files

Check whether the files are correctly placed:

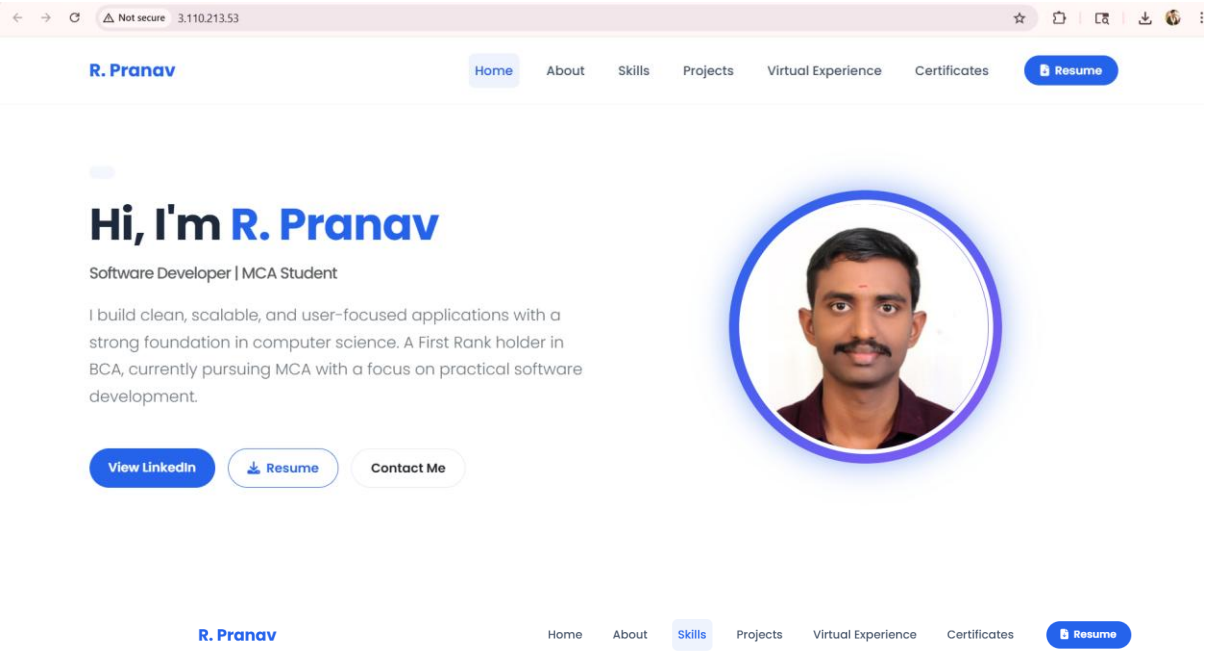
```
ls /var/www/html
```

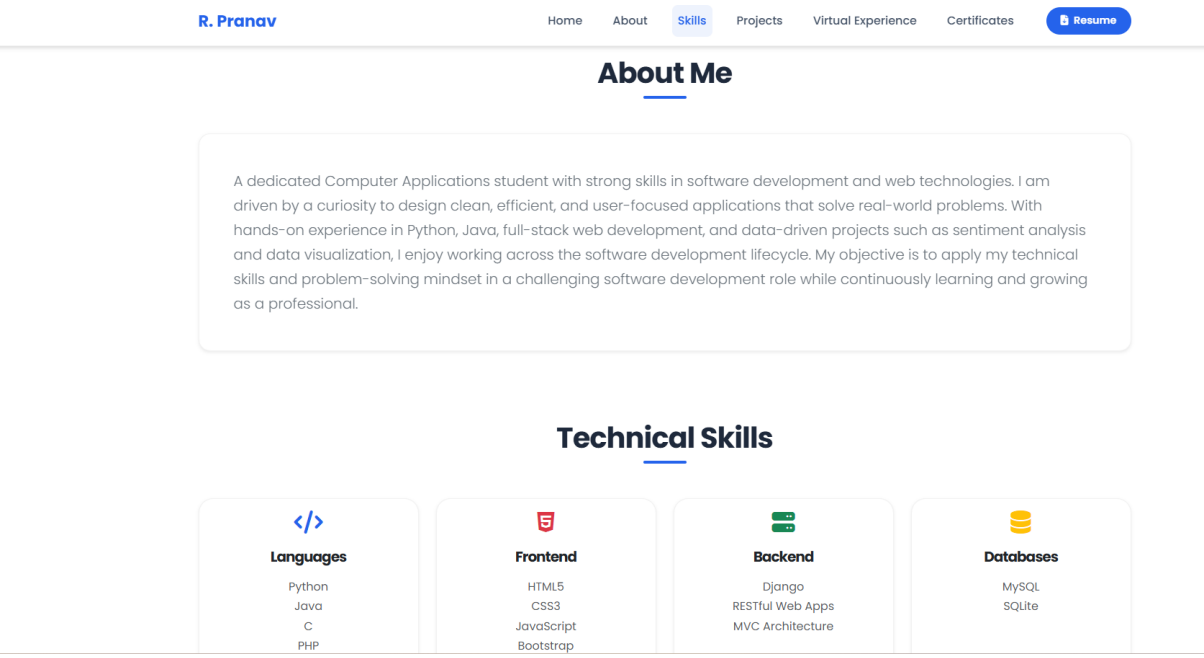
```
ubuntu@ip-172-31-38-145:~/MyPortfolio$ ls /var/www/html
assets  index.html  style.css
ubuntu@ip-172-31-38-145:~/MyPortfolio$
```

Step 9: Access Hosted Website

Open a web browser and enter:

http://3.110.213.53


Not secure 3.110.213.53
R. Pranav
Home
About
Skills
Projects
Virtual Experience
Certificates
Resume
Hi, I'm R. Pranav
Software Developer | MCA Student
I build clean, scalable, and user-focused applications with a strong foundation in computer science. A First Rank holder in BCA, currently pursuing MCA with a focus on practical software development.
View Linkedin
Resume
Contact Me


R. Pranav
Home
About
Skills
Projects
Virtual Experience
Certificates
Resume
About Me
A dedicated Computer Applications student with strong skills in software development and web technologies. I am driven by a curiosity to design clean, efficient, and user-focused applications that solve real-world problems. With hands-on experience in Python, Java, full-stack web development, and data-driven projects such as sentiment analysis and data visualization, I enjoy working across the software development lifecycle. My objective is to apply my technical skills and problem-solving mindset in a challenging software development role while continuously learning and growing as a professional.
Technical Skills
Languages
Python
Java
C
PHP
Frontend
HTML5
CSS3
JavaScript
Bootstrap
Backend
Django
RESTful Web Apps
MVC Architecture
Databases
MySQL
SQLite

Projects

ComplaintEase

Sentiment Analysis Web Application

ComplaintEase is a web-based complaint management system that automatically classifies user complaints into Positive, Negative, and Neutral categories using Natural Language Processing (NLP) techniques.

Key Features

- Analyze and classify sentiments
- User & Admin dashboards
- Data visualization insights
- Priority-based monitoring

Tech Stack

- Python & NLP
- HTML/CSS
- JavaScript
- Data Analytics

GitHub Repo

Achievements

First Rank Holder – Bachelor of Computer Applications (BCA)

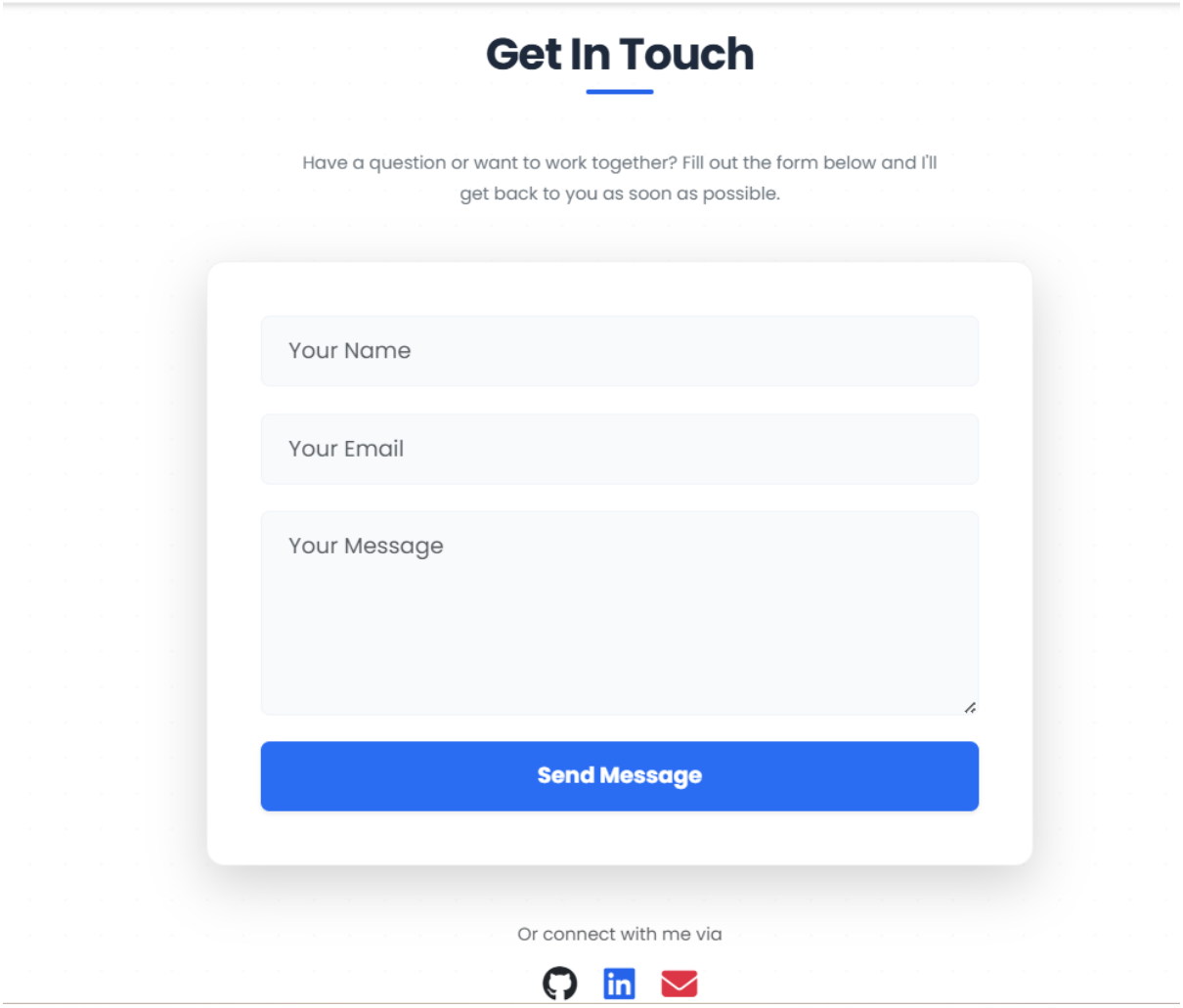
Secured First Rank in the BCA program under the University of Kerala, demonstrating consistent academic excellence and a strong understanding of core computer science concepts, programming, and software development fundamentals.

Certificates & Learning

HackerRank

Python Certification

Demonstrated foundational proficiency in Python programming, including data types, control structures, functional programming, and algorithmic problem-solving. This certification validates my ability to write clean and efficient Python code.



Step 10: Result

The custom website was hosted on an Ubuntu EC2 instance using Apache web server, and the files were transferred securely using SCP.